Platform	Description
TensorFlow.js	This is Google's open source initiative released in March 2018. So far, it's the best in the breed. It consists of various components. TensorFlow.js supports WebGL, a browser interface of OpenGL. This means that if the GPU is available, it will accelerate using it, otherwise it will fall back to the CPU. More details are available at *https://github.com/tensorflow/tfjs*
Keras.js	The semantic is exactly similar to that of TensorFlow in Python. More details at *https://github.com/transcranial/keras-js*
WebDNN	This is from the University of Tokyo. It is the fastest DNN execution framework in a browser. Details are at *https://github.com/mil-tokyo/webdnn*
ConvNetJS	ConvNetJS is a JavaScript library for training deep learning models (neural networks) entirely in your browser. It is no longer maintained. Details are at *https://cs.stanford.edu/people/karpathy/convnetjs/*
Brain.js	This is a 'GPU accelerated neural network in JavaScript for browsers and Node.js'. Details at *https://github.com/BrainJS*
Mind	This is a flexible neural network library for Node.js and the browser (*https://github.com/stevenmiller888/mind*).
Synaptic	Synaptic is a JavaScript neural network library for Node.js and the browser. Its generalised algorithm is architecture-free, so you can build and train basically any type of first order or even second order neural network architectures *(https://github.com/cazala/synaptic).*

Table 1: Deep learning browser open source frameworks

used to get the result of the localised test data set.

An example

Let us look at a very simplified JavaScript example of TensorFlow.js in action. The total code looks something like this:

```
<!DOCTYPE html>
<html>
    <head>
            <script src="https://cdn.jsdelivr.net/npm/@
tensorflow/tfjs/dist/tf.min.js"> </script>
    </head>
    <body>
         <h2>TensorFlow JS example. Input a number and get
prediction near to (3*input)-1.</h2>
        <p>
          <span>
          <label>Enter an integer here</label>
          <input type = "text"
                 id = "inputNumber"
                 placeholder = "Enter a number"
                value = "" />
          <label>(Beyond 1 to 10)</label>
          </span>
        </p>
        <button onclick="do_prediction()">Predict</button>
        <p id="prediction">Predicted Value: </p>
        <script>
            async function do_prediction(){
               const model=tf.sequential();
               model.add(
                   tf.layers.dense({
                       units:1,
                       inputShape:[1],
                       bias: true
                   })
               );
               model.compile({
                   loss:'meanSquaredError',
                   optimizer: 'sgd',
                   metrics: ['mse']
               });
               const train_labels = tf.tensor1d([1, 2, 3, 4,
5, 6, 7, 8, 9, 10]);
               const train_output = tf.tensor1d([2, 5, 8, 12,
14, 18, 21, 23, 26, 29]);
               await model.fit(train_labels, train_output,
{epochs:100});

               const inputVal = parseInt(document.
getElementById('inputNumber').value);
               document.getElementById('prediction').
innerText += model.predict(tf.tensor1d([inputVal])).
dataSync();
            }
        </script>
```